DEEP LEARNING AND CNN IMAGE CLASSIFICATION USING STREAMLIT

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1. Introduction

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. Machine Learning (ML) is a subset of AI that allows systems to learn from data and improve automatically without explicit programming. Deep Learning (DL) is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers to process complex data-andsolve-advanced-problems.

Deep Learning is widely used in image recognition, speech recognition, natural language processing, medical diagnosis, recommendation systems, robotics, and autonomous vehicles.

2. Difference Between AI, ML, and DL

Artificial Intelligence makes machines intelligent and capable of performing human-like tasks. Machine Learning allows systems to learn patterns from data automatically. Deep Learning uses multiple neural network layers to process complex data and solve advanced problems with high accuracy.

3. Applications of Deep Learning

- Image Recognition
- Speech Recognition
- Medical Diagnosis
- Self-Driving Cars
- Recommendation Systems
- Natural Language Processing
- Chatbots and Virtual Assistants

4. Types of Deep Learning Networks

The major types of Deep Learning Networks are:

1. Artificial Neural Network (ANN)
2. Convolutional Neural Network (CNN)
3. Recurrent Neural Network (RNN)
4. Long Short-Term Memory (LSTM)
5. Generative Adversarial Network (GAN)

5. Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is the basic form of neural network inspired by the human brain. ANN consists of neurons connected together in layers.

Components of ANN:
- Input Layer
- Hidden Layer
- Output Layer
- Weights
- Bias
- Activation Functions

Working of ANN:
1. Input data is provided to the input layer.
2. Data passes through hidden layers.
3. Weights and biases are applied.
4. Activation functions generate output.
5. Error is minimized using backpropagation.

Advantages:
- Learns complex patterns
- Improves prediction accuracy

Disadvantages:
- Requires high computational power
- Training takes more time

6. Convolutional Neural Network (CNN)

Convolutional Neural Network (CNN) is a deep learning model mainly used for image processing and computer vision tasks. CNN automatically extracts features from images without manual feature engineering.

Applications of CNN:
- Image Classification
- Face Recognition
- Object Detection
- Medical Image Analysis

7. CNN Architecture

The architecture of CNN consists of multiple layers:

1. Input Layer
Receives image data in the form of pixel values.

2. Convolution Layer
Extracts important features such as edges and textures using filters.

3. ReLU Activation Function
Introduces non-linearity into the model.
Formula:
$f(x) = \max(0, x)$

4. Pooling Layer
Reduces the size of feature maps while preserving important information.

5. Flatten Layer
Converts 2D feature maps into a 1D vector.

6. Dense Layer
Performs classification using extracted features.

7. Output Layer
Produces the final prediction using Softmax activation.

8. Working of CNN

1. Input image is given to the CNN model.
2. Convolution layer extracts features.
3. ReLU activation introduces non-linearity.
4. Pooling reduces dimensions.
5. Flatten converts data into vector format.
6. Dense layers perform classification.
7. Output layer predicts the final class.

9. Advantages and Disadvantages of CNN

Advantages:
• High accuracy in image tasks
• Automatic feature extraction
• Reduces manual work

Disadvantages:
• Requires large datasets
• High computational cost
• Longer training time

10. Recurrent Neural Network (RNN)

Recurrent Neural Network (RNN) is a neural network designed for sequential data processing. RNN remembers previous information using internal memory.

Applications:
• Speech Recognition
• Text Prediction
• Chatbots
• Language Translation

Advantages:
• Handles sequential data
• Maintains memory of previous inputs

Disadvantages:
• Slow training
• Vanishing gradient problem

11. CNN Image Classification Project

In this project, a CNN model was implemented for image classification using TensorFlow and Keras. The model was trained using an image dataset and deployed using Streamlit.

The project includes:
• CNN model implementation
• Image classification
• Model training
• Streamlit deployment
• Prediction display

12. Technologies Used

• Python
• TensorFlow
• Keras

- Streamlit
- NumPy
- Pillow
- OpenCV

13. Conclusion

Deep Learning is one of the most advanced technologies in Artificial Intelligence. ANN, CNN, and RNN are important deep learning architectures used for solving different types of problems. CNN provides excellent performance in image classification and computer vision tasks. In this project, CNN was successfully implemented and deployed using Streamlit for image classification.